

Joint segmentation of retinal layers and fluid lesions in optical coherence tomography with cross-dataset learning

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ABSTRACT

Background and objectives: Age-related macular degeneration (AMD) is the leading cause of irreversible vision loss among people over 50 years old, which manifests in the retina through various changes of retinal layers and pathological lesions. The accurate segmentation of optical coherence tomography (OCT) image features is crucial for the identification and tracking of AMD. Although the recent developments in deep neural network have brought profound progress in this area, accurately segmenting retinal layers and pathological lesions remains a challenging task because of the interaction between these two tasks.

Methods: In this study, we propose a three-branch, hierarchical multi-task framework that enables joint segmentation of seven retinal layers and three types of pathological lesions. A regression guidance module is introduced to provide explicit shape guidance between sub-tasks. We also propose a cross-dataset learning strategy to leverage public datasets with partial labels. The proposed framework was evaluated on a clinical dataset consisting of 140 OCT B-scans with pixel-level annotations of seven retinal layers and three types of lesions. Additionally, we compared its performance with the state-of-the-art methods on two public datasets.

Results: Comprehensive ablation showed that the proposed hierarchical architecture significantly improved performance for most retinal layers and pathological lesions, achieving the highest mean DSC of 76.88 %. The IRF also achieved the best performance with a DSC of 68.15 %. Comparative studies demonstrated that the hierarchical multi-task architecture could significantly enhance segmentation accuracy and outperform state-of-the-art methods.

Conclusion: The proposed framework could also be generalized to other medical image segmentation tasks with interdependent relationships.

1. Introduction

Age-related macular degeneration (AMD) is the leading cause of irreversible vision loss in the elderly population of industrialized nations [1]. AMD primarily affects the macula, which is the cone-photoreceptor-rich central part of the retina. Early AMD is characterized by lipoprotein-rich deposits called drusen. Advanced AMD is classified as neovascular (or 'wet') and atrophic (or 'dry') AMD, with the former characterized by proliferative neovascularization and the latter characterized by atrophy

of the retinal pigment epithelium and the outer neurosensory retina. Optical coherence tomography (OCT) provides high-resolution cross-sectional imaging of the retina and is widely used in disease diagnosis, treatment-decision guidance, and therapy-response assessment. The accurate quantification of OCT image features is thus crucial for the identification and tracking of AMD.

The image features of AMD are highly complex, including changes in retinal layers and various types of pathological lesions [2]. First, retinal layers have a strict anatomical ordering and are separated by gradual

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gradients. The task is further complicated by the fact that certain retinal layers may undergo atrophy in specific diseases (e.g., atrophy of retinal pigment epithelium or neurosensory retina). Second, the location of retinal layers and lesions can influence each other. Lesions are confined by the topology of retinal layers, while the presence of lesions can also alter the shape of retinal layers. The accurate segmentation of retina layers and pathological lesions in OCT remains a highly challenging task.

Recent developments in deep neural networks have brought profound progress in medical image analysis. Several studies have proposed simultaneous segmentation of retinal layers and fluid lesions, with some of them incorporating constraints between these two tasks (see Section 2.1 for details). However, these methods impose simple one-way constraints, which are unable to capture the mutual interaction between retinal layers and fluid lesions. To address this limitation, we propose a novel two-way constraint strategy in this study, enabling the mutual influence and interaction between retinal layers and fluid lesions.

In this study, we propose an innovative three-branch, hierarchical multi-task framework that facilitates the mutual influence between retinal layers and fluid lesions. Specifically, a novel regression guidance module (RGM) is introduced to provide shape guidance between the branches. The proposed method was extensively evaluated on a clinical dataset of 140 OCT B-scans, which has pixel level annotation for seven retinal layers and three types of pathological lesions. We also compared the proposed method with state-of-the-art approaches on several widely used public datasets. These datasets typically focus on either layer segmentation or fluid segmentation individually. To address this limitation, we introduce a semi-supervised cross-dataset learning approach that leverages public datasets with partial labels, enabling more comprehensive learning. To summarize, our main contributions in this paper are as follows:

1. Hierarchical multi-task architecture: we propose a three-branch, hierarchical multi-task framework, allowing for the joint regression of seven retinal layers and segmentation of three pathological lesions. A regression guidance module is introduced to provide explicit shape guidance between sub-tasks.
2. Cross-dataset learning: we propose a cross-dataset learning strategy based on pseudo-label to leverage public datasets with partial labels.
3. Extensive evaluation: the proposed method was extensively evaluated through ablation and comparative studies on a real-world clinical dataset and two public datasets. The results showed that the proposed framework outperforms state-of-the-art methods across a range of metrics. The clinical dataset, manual labels, and code are publicly available on GitHub (<https://github.com/xjtu-mia/OCT-Multitask>).

2. Related work

2.1. Segmentation of retinal layers and fluid lesions in OCT

The segmentation of retinal layers and lesions is usually treated as separate tasks. In this part, we first review the recent works in layer segmentation, and then in lesion segmentations. At last, we report the studies that jointly deal with layers and lesions.

Retinal layers have strict anatomical ordering, with adjacent layers separated by vague gradients known as layer-surfaces. Thus, a retinal layer can be directly detected as a region with certain thickness, or be defined by its upper and lower layer-surfaces. Ngo et al. formulated the layer-surface detection task as a regression problem by predicting the height of the surface [3]. Mukherjee et al. proposed a 3D convolutional regression network to predict layer-surfaces [4]. These methods cannot guarantee the topological ordering of layers. Towards this, He et al. proposed a spatial constraint mechanism to guarantee the correct topological relationships between retinal layers [5]. Some other studies work towards reducing the high dependency on labeled data. For

example, Fazekas et al. proposed a semi-supervised layer segmentation using disentangled representation and anatomical priors to segment retinal layers [6].

AMD related pathological fluid lesions mainly include intra-retinal fluid (IRF), sub-retinal fluid (SRF), and pigment epithelial detachment (PED). IRFs are hypo-reflective regions between the inner limited membrane (ILM) and the outer nuclear layer (ONL). SRFs are hypo-reflective regions between the outer boundary of neurosensory retina and the inner boundary of retinal pigment epithelium (RPE). PED is defined as the detachment of the retinal pigment epithelium layer from the Bruch's membrane (BM). Significant effort has been made to develop automated methods for the segmentation of single type/multiple types of fluid lesions in OCT. The latest developments mainly take advantage of the deep learning techniques [7–12]. In the RETOUCH Grand Challenge, all nine participant teams used deep learning methods, with most of them implementing a variant of fully convolutional network (FCN) [11]. A recent work by Xing et al. reported a novel curvature loss function to simultaneously segment three types of fluid lesions [12].

Although retinal layer segmentation and fluid lesion segmentation can be addressed separately, they are not independent tasks. As mentioned, retinal layers and lesions can interact with each other, and ignoring this relationship may result in unrealistic segmentations. Several studies have proposed to simultaneously segment retinal layers and fluid lesions by modeling the problem either as: 1) a multi-class segmentation problem, or 2) a sequential segmentation problem using cascaded models. Segmentation constraints are provided from layers to fluid lesions [13–15] or between layers and fluid lesions [16,17]. However, these methods impose one-way constraints, meaning a subsequent task cannot influence a preceding task. In contrast, we propose a two-way constraint strategy in this study, which allows for mutual influence between retinal layers and fluid lesions.

2.2. Multi-task learning and its applications in medical image analysis

Multi-task learning (MTL) refers to a general learning approach where multiple tasks are handled simultaneously. MTL has been successfully applied in various domains, such as natural language processing, bioinformatics, and computer vision tasks. In the era of deep learning, MTL is typically in the form of a shared encoder that branches into several task-specific heads [18]. MTL can be further classified into encoder-focused architectures and decoder-focused architectures. The former decodes each task with an independent task-specific head, while the latter allows various forms of information exchange between tasks during the decoding stage. Specifically, it is argued that not all tasks always benefit from joint learning, nor do all tasks have equivalent importance. Along this line, asymmetric MTL is proposed, which is designed to allow asymmetric information transfer between the tasks [19].

Various forms of MTL architectures have been applied for medical image analysis. For example, Amyar et al. proposed a multi-task deep learning approach for joint classification and segmentation of COVID-19 pneumonia, in which three independent task-branches are designed for segmentation, classification, and reconstruction [20]. Cheng et al. proposed a MTL framework for simultaneous glima segmentation and IDH genotyping with two independent task branches [21]. Focusing on the issue that prostate bed segmentation is critically dependent on the neighboring bladder and rectum segmentation, Xu et al. proposed a two-branch asymmetric multi-task attention network that considered the one-way dependency between prostate bed segmentation and bladder segmentation [22]. In this study, we extend previous work and provide a novel three-branch multi-task architecture that allows two-way dependency between the retinal layer regression task and pathological lesions segmentation task.

3. Methods

3.1. Preprocessing

A three-dimensional block matching filtering is applied to reduce noise, followed by a Sobel filtering to generate a gradient map image. The top and bottom boundaries of the retina (i.e., ILM and BM) are identified based on the intensity of the gradient, and a convex hull fitting is performed to correct potential errors in the BM boundary. Retina flattening is applied by adjusting the BM height to a horizontal line, and the region of interest (ROI) is cropped from 80 pixels (0.31 mm) below the BM to 272 pixels (1.07 mm) above, resulting in a 352×512 pixel image. At last, intensity normalization is applied by rescaling the contrast on each B-scan.

3.2. Three-branch, hierarchical multi-task architecture

The proposed multi-task architecture is shown in Fig. 1. We propose a three-branch, hierarchical architecture consisting of two layer-surface-regression (LSR) tasks and one pixel-classification (PC) task. The attention blocks (AB) allow for asymmetric information flow between sub-tasks, generating a hierarchy between sub-tasks.

3.2.1. Encoder-decoder structure

The proposed framework has one encoder and three decoders. The encoder is designed based on a pre-activation ResNet50, wherein we provide a number of task-specific modifications. First, similar as introduced in ConvNext, we adjust the number of residual blocks from [3,4,6] to [3,9] and remove the first two activation functions. Second, the ReLU activation functions are replaced with ACON activation functions, which can avoid the ‘dead neurons’ issue and allows the neurons to adaptively activate. Third, the original down-sampling strategy (i.e., convolution with stride 2) is replaced by convolution with stride 1 plus max-pooling, which is more effective in preserving the gradient information for layer-

surface segmentations.

The three decoders can be divided into two groups. Decoder-1 is similar to the decoder of a vanilla U-Net, consisting of four Convolutional blocks (CBs) that include a double convolutional layer and an up-sampling layer. Other minor modifications are proposed, including replacing the transpose convolution in up-sampling with nearest neighbor interpolation with 2×2 convolution and replacing the ReLU activation functions with GELU.

Decoder-2 and 3 apply multiple attention blocks, which allows soft constraints in the form of attention weight from a lower decoder branch to a higher decoder branch (Fig. 2). Taking decoder-3 for example, the feature maps x_{i-1} output by $AB_{d3,i-1}$ is first fed into an up-sampling module to double the resolution of the feature maps:

$$x_{up,i} = \mathcal{N}(\mathcal{F}_{2 \times 2}(x_{i-1}))_{\times 2} \quad (1)$$

where $\mathcal{F}_{2 \times 2}(\bullet)$ denotes the operation of [Conv2d 2×2 -BN-GELU], $\mathcal{N}(\bullet)_{\times 2}$ denotes nearest interpolation up-sampling.

Then $x_{up,i}$ and y_i output by $AB_{d2,i}$ are concatenated as $x_{cat,i} = \mathcal{C}(x_{up,i}, y_i)$, where $\mathcal{C}(\bullet)$ denotes the channel-wise concatenation operation. On the basis of $x_{cat,i}$, the computation of generating attention mask can be express as follows:

$$a_i = \sigma(\mathcal{F}'_{1 \times 1}(\mathcal{F}_{1 \times 1}(x_{cat,i}))) \quad (2)$$

where $\mathcal{F}_{1 \times 1}(\bullet)$ denotes [Conv2d 1×1 -BN-GELU], $\mathcal{F}'_{1 \times 1}(\bullet)$ denotes [Conv2d 1×1 -BN], and σ is the sigmoid function. Finally, we can get the output of $AB_{d3,i}$:

$$x_i = \mathcal{F}_{3 \times 3}(x_{cat,i} \otimes a_i) \quad (3)$$

where $\mathcal{F}_{3 \times 3}(\bullet)$ denotes [Conv2d 3×3 -BN-ACON] and $\otimes$ denotes element-wise multiplication.

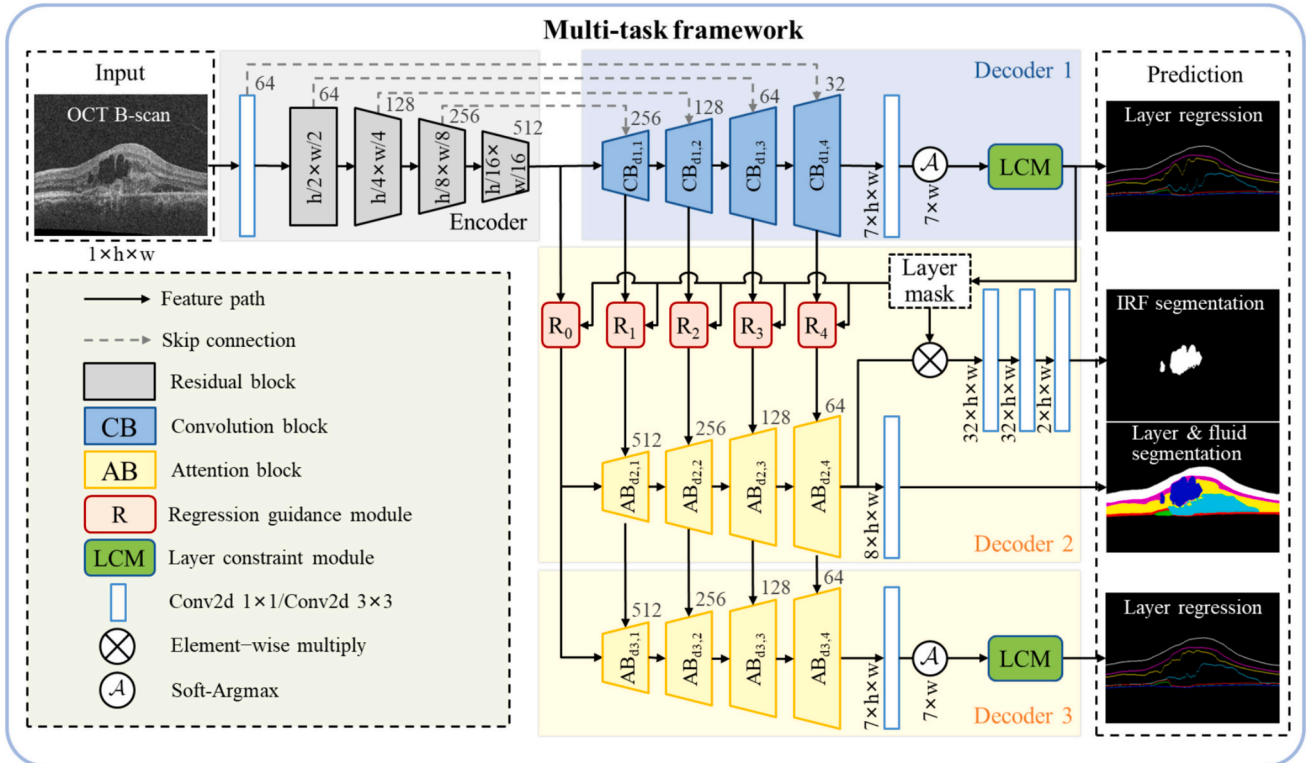


Fig. 1. An overview of the multi-task framework. The proposed multi-task framework is a three-branch, hierarchical architecture consisting of two LSR tasks and one PC task. Asymmetric information flow exists between sub-tasks through attention blocks.

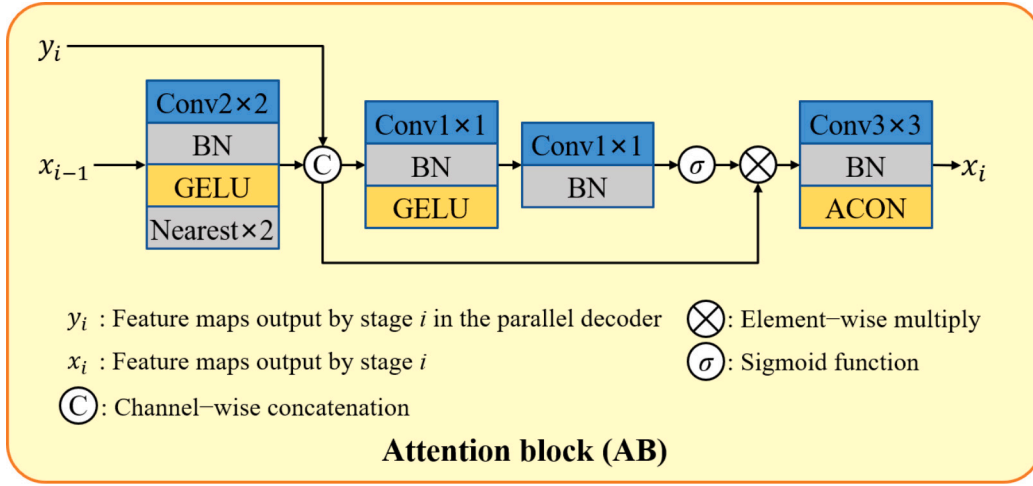


Fig. 2. The structure of the attention block.

3.2.2. Layer-surface regression subtask

Layer regression, which directly predicts layer-surface locations in each image column, ensures the continuity of layers and has been used for layer segmentation in previous studies. Given an $X \in \mathbb{R}^{h \times w}$, where h and w are the height and width of X , a layer-surface can be regarded as a curve running through the image along the x-axis. Thus, in each column, there must have one pixel belonging to the layer-surface. On this basis, we adapt the coordinate estimation methods widely used in human pose estimation and keypoint estimation. A given output feature map X can be split into w independent column vectors with a width of 1 pixel:

$$X = [r_0, r_1, \dots, r_n, \dots, r_{w-1}] \quad (4)$$

where $r_n = [r_n^0, r_n^1, \dots, r_n^{h-1}]^T$ denotes the n^{th} column vector. The position of the layer-surface in each column follows an independent distribution and a Softmax function is applied to each column vector to map the output to 0–1:

$$P_n = \text{softmax}(\beta r_n) = \frac{\exp(\beta r_n)}{\sum_{i=0}^{h-1} \exp(\beta r_n^i)} \quad (5)$$

where β is an adjustment factor that controls the smoothness of probability distribution. We define that the surface position in column r_n is s_n whose expectation can be calculated as:

$$(s_n) = \text{id}x^T \cdot P_n \quad (6)$$

where $\text{id}x = [0, 1, \dots, h-1]^T$ indicates the index of pixels in a column vector.

A model predicting L layer-surfaces contains L feature maps, with each feature map containing the prediction of a single layer-surface. In this scenario, if no interaction exists between these feature maps, the anatomical ordering of these predicted layers cannot be explicitly guaranteed. Thus, we add a layer constraint module (LCM) to ensure the correctness of the topological ordering between layers similar as in [5]. Taking the n^{th} column for example, for the j^{th} surface $s_{j,n}$ and $j-1$ -th surface $s_{j-1,n}$ (the smaller the j , the closer it is to the inner retina, with $s_{0,n}$ denotes a position on ILM), the final prediction of $s_{j,n}$ can be calculated as:

$$s_{j,n} = s_{j-1,n} + \text{Softplus}(s_{j,n} - s_{j-1,n}) \quad (7)$$

where Softplus can be regarded as the smoothing approximation of ReLU. We choose Softplus because it is continuous in its definition domain and the output value is always greater than zero. The LCM explicitly guarantees a correct anatomical ordering of the predicted layer-surfaces.

3.2.3. Pixel-classification subtask

Decoder-2 is a pixel-classification subtask, which segments the fluid lesions that cannot be determined in the previous regression subtask (decoder-1) and provides important feature maps for layer refinement in the last regression subtask (decoder-3). Among pathological lesions, IRFs can span multiple layers, whereas SRFs and PEDs are typically confined by continuous layers. To better address the unique characteristics of IRFs, a dedicated IRF branch (IB) is introduced to enhance performance. Specifically, feature maps before the last block are multiplied by a binary layer mask that identifies the region between ILM and BM, which is then fed into a distinct IRF output branch to independently optimize the IRF segmentation task.

3.2.4. Regression guidance module

To provide more guidance between subtasks, we introduce a regression guidance module (Fig. 3). RGM is an attention gate mechanism, consisting of a hard spatial attention and a channel attention. First, layer region masks are generated from the output of decoder-1, which are used as a hard spatial attention [23]. Specifically, spatial attention updates the feature maps at each position by multiplying the mask with feature maps, allowing an explicit shape guidance for each layer. The second part of RGM is a learnable feature filtering module (FFM), which is an efficient channel attention [24]. FFM allows the model to adaptively pay attention to specific feature channels.

Specifically, the predicted 1-D layer-surfaces are converted into mutually exclusive 2-D layer masks, which are then resized to the size of each decoder level and element-wise multiplied with the corresponding feature maps from decoder-1. Taking the last RGM block (i.e., R4) for example:

$$[x_{m1}, x_{m2}, \dots, x_{mj}, \dots, x_{mc-1}] = x_{d1,4} \otimes [m_1, \dots, m_j, \dots, m_{c-1}] \quad (8)$$

where $x_{d1,4}$ denotes the feature maps from the 4th level of decoder-1, m_j is the j^{th} 2-D layer mask, c is the number of layer surfaces, $\otimes$ denotes element-wise multiplication, x_{mj} denotes the output feature maps processed by the j^{th} layer mask. In this way, a hard attention is provided from decoder-1 to decoder-2.

Then, a learnable feature filtering module (FFM) is used to re-weight the feature channels. The j^{th} FFM can be described as:

$$g_j = \mathcal{F}_{3 \times 3} \left(x_{mj} \otimes \sigma \left(\mathcal{F}_k \left(\mathcal{G}(x_{mj}) \right) \right) \right) \quad (9)$$

where $\mathcal{G}(\bullet)$ denotes channel-wise global average pooling (GAP), $\mathcal{F}_k(\bullet)$ denotes 1-D convolutional layer with a kernel size of k , σ denotes sigmoid function, and $\mathcal{F}_{3 \times 3}(\bullet)$ denotes [Conv2d 3 × 3–BN–ACON], respectively. Finally, the original feature maps and all of feature maps

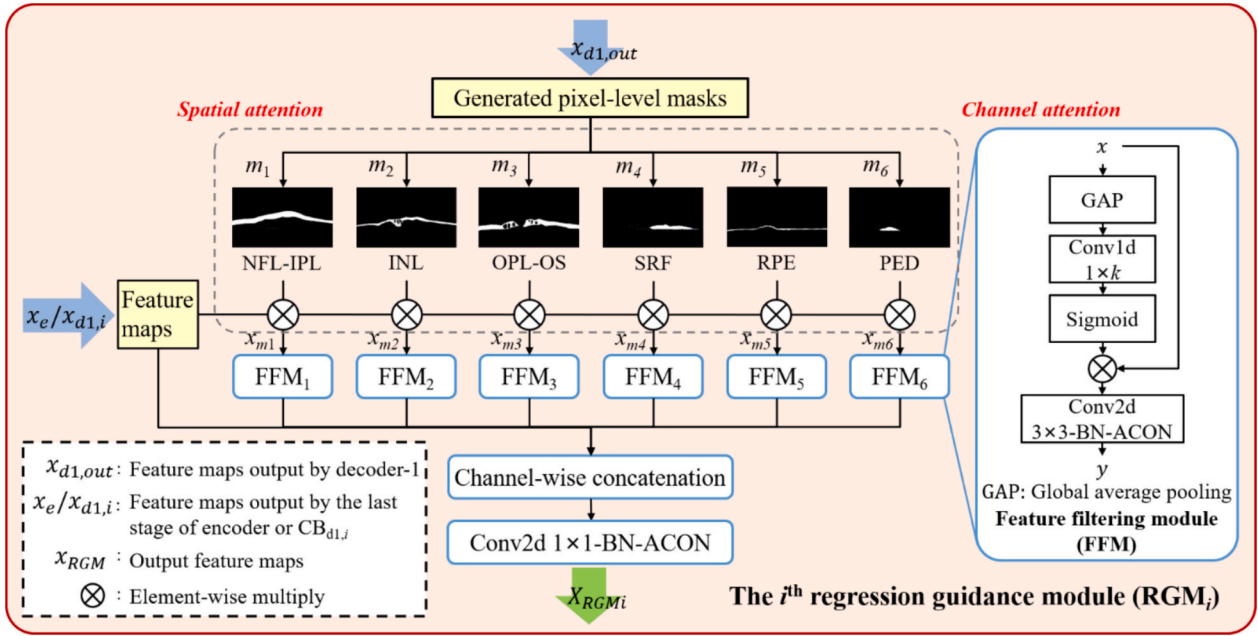


Fig. 3. The structure of the regression guidance module.

output by FFMs are merged:

$$x_{RGM_4} = F_{1 \times 1} \left(C \left(x_{d1,4}, g_1, g_2, \dots, g_i, \dots, g_{c-1} \right) \right) \quad (10)$$

where $C(\bullet)$ the channel-wise concatenation operation, $F_{1 \times 1}(\bullet)$ denotes $[Conv2d \ 1 \times 1-BN-ACON]$, and x_{RGM_4} is the output feature maps of the last RGM.

3.3. Semi-supervised learning across datasets

The proposed architecture is evaluated on public datasets, including Duke and RETOUCH. However, RETOUCH only contains manual labels for fluid lesions and does not have labels for retinal layers. Thus, to make use of the RETOUCH dataset, we adopt a cross-dataset learning strategy

based on pseudo-label to solve the problem of missing labels in the RETOUCH dataset (Fig. 4).

In step-1, a model is trained with the fully labeled clinical dataset (source dataset). In step-2, the trained model is used to process the RETOUCH dataset (target dataset) to obtain a prediction containing all labels. In step-3, the predicted labels are corrected based on the manual labels. Specifically, predicted fluid lesion labels are replaced by the fluid lesion labels provided by manual graders. Then, the predicted retinal layers are adjusted according to their topological relationship with the fluid lesions. For example, the outer boundary of neuroretina and the inner boundary of RPE are adjusted according to their topological relationship with SRF. The outer boundary of RPE and BM are adjusted according to their topological relationship with PED. The other retinal layers are also adjusted if a topological conflict exists.

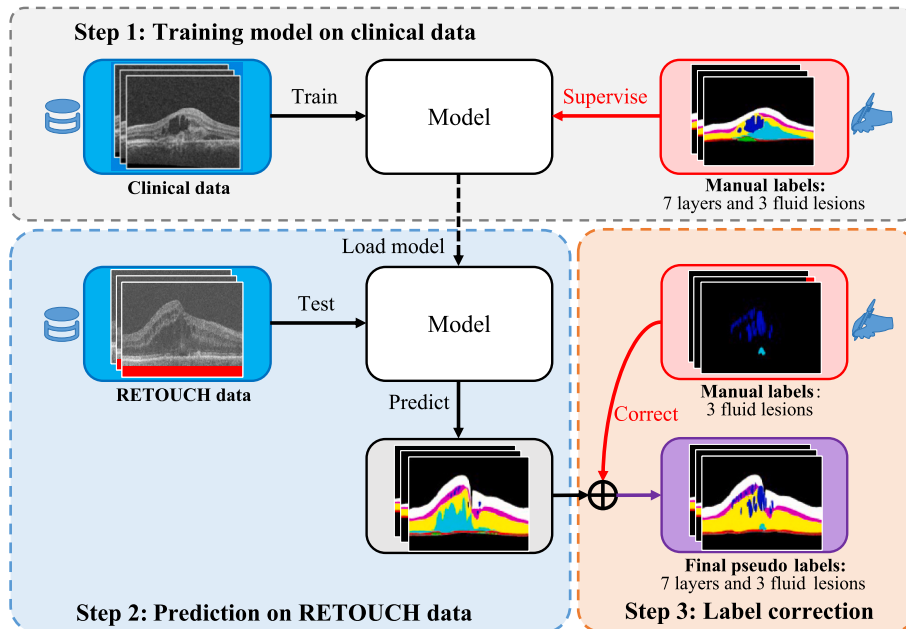


Fig. 4. Illustration of the pseudo-label generation strategy.

3.4. Loss function

The loss function of our proposed model is a combination of regression loss (decoder-1 and decoder-3) and classification loss (decoder-2). First, for an image of size $H \times W$ pixels and M classes, the regression loss consists of a column-wise cross-entropy loss $\mathcal{L}_{CE_col}$ and a smooth-L1 loss $\mathcal{L}_{sL1}$, which are defined as:

$$\mathcal{L}_{CE_col} = \frac{1}{MW} \sum_{i \in \{M, W\}} -g_H(i)^T \bullet \ln(p_H(i)) \quad (11)$$

$$\mathcal{L}_{sL1} = \frac{1}{MW} \sum_{i \in \{M, W\}} \text{smooth}_{L1}(idx^T \bullet p_H(i) - idx^T \bullet g_H(i)), (idx^T = [0, 1, \dots, H-1]) \quad (12)$$

$$\text{smooth}_{L1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases} \quad (13)$$

where $p_H(i)$ and $g_H(i)$ denote the predicted position vector and ground-truth position vector along each column. Considering that BM is prone to segmentation error in the case of large PED, we design a Laplacian regularization (LR) term to increase the penalty for an uneven BM prediction. Thus, the final regression loss can be expressed as:

$$\mathcal{L}_{\text{regression}} = \mathcal{L}_{CE_col} + \mathcal{L}_{sL1} + \alpha \frac{1}{W} \|\nabla_k^2 * p_{BM}\|_1 \quad (14)$$

where p_{BM} denotes the row vector composed of the position expectation of BM, ∇_k^2 is a one-dimensional Laplace convolution operation with a width of k , $\|\cdot\|_1$ denotes the L1 norm, and α is a scale factor of the regularization term. The width k of Laplace convolution operation should be able to capture the unevenness of the BM. In our experiments, we set $k = 51$ and $\alpha = 0.2$.

Second, the classification loss consists of a cross-entropy loss $\mathcal{L}_{WCE}$ and a Dice loss $\mathcal{L}_{Dice}$, which are defined as:

$$\mathcal{L}_{WCE} = \frac{1}{HW} \sum_{i \in \{H, W\}} \sum_{c=1}^M -g_c(i) \ln(p_c(i)) \quad (15)$$

$$\mathcal{L}_{Dice} = 1 - \frac{1}{M} \sum_{c=1}^M \frac{2 \sum_{i \in \{H, W\}} p_c(i) g_c(i)}{\sum_{i \in \{H, W\}} (p_c(i) + g_c(i)) + \varepsilon} \quad (16)$$

where $p_c(i)$ is the probability of pixel i belonging to class c , $g_c(i)$ is 1 for pixels belonging to class c and 0 for other pixels, ε is a small constant used to prevent division by zero. In our experiments, we set $\varepsilon = 1e-7$. Thus, the final classification loss can be expressed as:

$$\mathcal{L}_{\text{classification}} = \mathcal{L}_{CE} + \beta \mathcal{L}_{Dice} \quad (17)$$

where β is the scale factor of Dice loss. The parameter β balances the two loss functions and β is 0.4 in our experiment. At last, the total loss of the proposed model is:

$$\mathcal{L} = \mathcal{L}_{\text{regression1}} + \mathcal{L}_{\text{regression2}} + \lambda (\mathcal{L}_{\text{classification1}} + \mathcal{L}_{\text{classification2}}) \quad (18)$$

where $\mathcal{L}_{\text{regression1}}$ and $\mathcal{L}_{\text{regression2}}$ denote the loss functions for decoder-1 and decoder-3, $\mathcal{L}_{\text{classification1}}$ denotes the loss function for decoder-2, $\mathcal{L}_{\text{classification2}}$ denotes the loss function for the IRF branch, and λ is a

scale factor between regression task and classification task.

4. Experiments

4.1. Materials

4.1.1. Clinical dataset

The clinical dataset consisted of 50 OCT volumes from 50 neovascular AMD patients at the First Affiliated Hospital of Xi'an Jiaotong University. All OCT volumes were acquired with Cirrus HD 4000 (size of $512 \times 1024 \times 128$ pixels and an axial resolution of $1.96 \mu\text{m}$). A set of 1–5 B-scan images were selected from each OCT volume, resulting in a total

of 140 B-scans. SRF are shown in 124 images, PED in 93 images, and IRF in 28 images. This study is in adherence to the tenets of the Declaration of Helsinki and was approved by the Institutional Review Board of Xi'an Jiaotong University.

Seven retinal layer-surfaces were manually annotated at pixel level by two graders and supervised by one ophthalmologist (Fig. 5), including ILM, inner plexiform layer-inner nuclear layer (IPL-INL), inner nuclear layer-outer plexiform layer (INL-OPL), outer surface of the photoreceptor cells (outerPRC), inner surface of retinal pigment epithelium (IRPE), outer surface of retinal pigment epithelium (ORPE), and BM. The contour of IRF was also manually annotated. SRF is defined as the region between outerPRC and IRPE, while PED is defined as the region between ORPE and BM.

4.1.2. Public datasets

The Duke dataset contains 110 OCT B-scans with pixel-level annotation from 10 diabetes macular edema patients [25]. OCT B-scans were acquired with Spectralis SD-OCT device with 512×728 pixels and axial resolutions of $3.87 \mu\text{m}$. The annotations were done by two experts, including IRF and eight retinal layer-surfaces. The RETOUCH dataset consists of 112 OCT volumes from 112 AMD and RVO patients [11]. The training set contains 24, 24 and 22 OCT volumes acquired with Cirrus, Spectralis and Topcon scanners, respectively. Pixel-level annotations of three types of fluid lesions (i.e., SRF, PED, and IRF) are provided by experts.

4.2. Evaluation metrics

For the LSR task, we use mean absolute distance (MAD) in μm and dice similarity coefficient (DSC) to evaluate the regression results for layer-surfaces. For PC task, we use DSC to evaluate the pixel-classification results for both layers and fluid lesions. These two metrics are defined as follows:

$$\text{MAD} = \frac{1}{W} \sum_{i=0}^{W-1} \text{abs}(X(i) - Y(i)) \quad (19)$$

$$\text{DSC} = \frac{2|P \cap G|}{|P| + |G|} \quad (20)$$

where $X(i)$ and $Y(i)$ are the predicted and true values of the row index in the i^{th} column, W is the total number of columns of a B-scan image, P and G represent the predicted segmentation, and the ground truth, respectively.

We also used absolute volume difference (AVD) in mm^3 to evaluate the differences between the predicted fluid segmentation and the ground

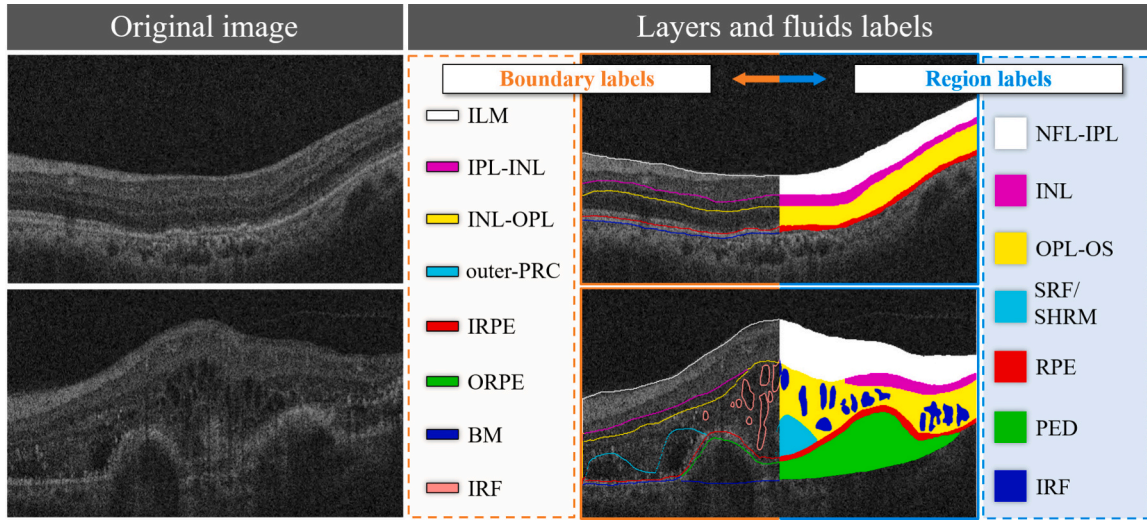


Fig. 5. Illustration of retinal layers and pathological fluid lesions in normal and AMD cases.

truth for the RETOUCH dataset:

$$AVD = \text{abs}(V_x(i) - V_y(i)) \quad (21)$$

where $V_x(i)$ and $V_y(i)$ denote the predicted volume and truth volume of the fluid in the i^{th} OCT volume, respectively.

4.3. Implementation details

The implementation of the proposed framework was based on Pytorch library. All experiments were conducted using a NVIDIA GeForce RTX3090. For training the proposed framework, we performed AdamW optimizer with initial weight decay of $1e-4$, initial learning rate of $1e-4$, β_1 and β_2 to the default values of 0.9 and 0.99, and batch size of 2. For all training processes, when the iteration times reached 90 % of the max iteration, we apply a poly decay strategy a power of 1.5 to the initial learning rate and initial weight decay, gradually reducing them to $1e-8$. The clinical data is split into training and test sets, with the 70 B-scans from the first 26 volumes as training and the 70 B-scans from the rest 24 volumes as test set. For the public Duke dataset, we divided the dataset into two even sets (i.e., patient 1–5 and patient 6–10) for cross-validation similar as the literatures. For the public RETOUCH dataset, six-fold cross validation was performed on each device.

5. Results

5.1. Ablation study for the three-branch architecture

First, we conducted the ablation study for the three-branch, hierarchical architecture on the clinical dataset. To do this, four models were designed and compared. MT1B-PC denotes the proposed decoder-1 with a PC output, and MT1B-LSR denotes decoder-1 with an LSR output. MT2B includes the proposed decoder-1 and decoder-2. At last, MT3B is

Table 1

Ablation study for the three-branch architecture in DSC (mean $\pm$ std) (%).

Class	MT1B-PC	MT1B-LSR	MT2B	MT3B
NFL-IPL	92.46 $\pm$ 0.15	92.51 $\pm$ 0.14	92.37 $\pm$ 0.13	92.68 $\pm$ 0.07
INL	74.65 $\pm$ 0.44	72.89 $\pm$ 0.26	74.32 $\pm$ 0.43	73.27 $\pm$ 0.41
OPL-OS	87.23 $\pm$ 0.26	87.27 $\pm$ 0.22	86.76 $\pm$ 0.24	87.59 $\pm$ 0.04
SRF	72.81 $\pm$ 0.59	72.05 $\pm$ 1.13	72.31 $\pm$ 0.70	74.83 $\pm$ 0.60
RPE	63.46 $\pm$ 0.18	62.79 $\pm$ 0.61	63.04 $\pm$ 0.49	63.22 $\pm$ 0.21
PED	67.92 $\pm$ 1.41	67.31 $\pm$ 1.37	67.30 $\pm$ 0.77	69.66 $\pm$ 0.99
IRF	62.60 $\pm$ 4.17	–	66.57 $\pm$ 3.89	68.15 $\pm$ 2.18
Mean	74.45 $\pm$ 0.60	75.80 $\pm$ 0.47	74.67 $\pm$ 0.48	75.63 $\pm$ 0.39

the proposed three-branch, hierarchical architecture. All models have the same encoder.

The results are shown in Table 1. Compared with MT1B-LSR, MT1B-PC showed a slightly superior performance in layer surface segmentation. MT1B-LSR cannot explicitly segment pathological lesions, thus the existence of pathological lesions may result in erroneous segmentations, as reflected by the lower DSC for SRF and PED. Compared with MT1B-PC, MT2B showed an obvious performance gain in IRF segmentation, with DSC from 62.60 % to 66.57 %. Meanwhile, MT2B showed a slight performance degradation for most layers, indicating that the layer surface regression task did not benefit from a higher-level PC task. At last, the proposed MT3B gave a significantly better performance for most retinal layers and pathological lesions. IRF also achieved the best performance with a DSC of 68.15 %. The results indicate that both the layer surface regression task and pathological lesion segmentation task benefit from the three-branch hierarchical architecture.

5.2. Ablation study of the proposed additional modules

In this section, we conducted the ablation studies for the proposed IB and RGM modules, and further for the Laplace regularization (LR) loss (Table 2) on the clinical dataset. Visualization of results is given in Fig. 6.

Starting from the baseline model, adding an independent IRF branch significantly improved the IRF segmentation result, with the DSC increased from 55.91 % to 61.99 %. Meanwhile, the performance of layer segmentation improved slightly, with the mean MAD decreased from 12.90 μm to 12.73 μm . The reason may be that a more accurate IRF location provided more accurate constraints for neighboring layers. Adding RGM to the model also improved the performance on most layers and fluid lesions, with the mean MAD decreased from 12.90 μm to 12.65 μm , and the mean DSC increased from 73.15 % to 74.77 %. The performance gain was even larger for the model with both IB and RGM, with the DSC increased by >12 % for IRF compared with the baseline.

Finally, the proposed LR loss significantly improved the accuracy of BM prediction, with the MAD significantly decreased from 12.57 μm to 11.84 μm . It is noteworthy that the segmentation of PED, which is topologically constrained by BM, also showed an increased DSC from 69.66 % to 70.42 %.

5.3. Comparative study of different encoders

In this section, we compared the proposed modified ResNet backbone with six other well-known classification models on the clinical

Table 2

Ablation study of the proposed additional modules and the LR loss. Paired student's *t*-test are performed between the first column and the other columns, with one and two asterisks denoting a significantly better performance with $p < 0.05$ and $p < 0.01$, respectively.

MAD (mean $\pm$ std) (μm)					
Class	Baseline	Baseline+IB	Baseline+RGM	Baseline+IB + RGM	(Baseline+IB + RGM) + LR loss
ILM	5.46 $\pm$ 0.14	5.42 $\pm$ 0.09	5.31 $\pm$ 0.06	5.34 $\pm$ 0.10	5.29 $\pm$ 0.06
IPL-INL	10.67 $\pm$ 0.16	10.54 $\pm$ 0.15	10.41 $\pm$ 0.19	10.47 $\pm$ 0.17	10.36 $\pm$ 0.12 *
INL-OPL	16.17 $\pm$ 0.49	16.37 $\pm$ 0.20	16.39 $\pm$ 0.67	15.98 $\pm$ 0.36	16.03 $\pm$ 0.38
OuterPRC	12.37 $\pm$ 0.44	11.94 $\pm$ 0.31	11.84 $\pm$ 0.28	12.03 $\pm$ 0.26	11.84 $\pm$ 0.26
IRPE	17.49 $\pm$ 0.33	17.54 $\pm$ 0.30	16.98 $\pm$ 0.68	16.68 $\pm$ 0.35 **	16.69 $\pm$ 0.14 **
ORPE	15.15 $\pm$ 0.22	14.76 $\pm$ 0.31	14.56 $\pm$ 0.60	14.47 $\pm$ 0.21 **	14.67 $\pm$ 0.14 **
BM	12.98 $\pm$ 0.55	12.52 $\pm$ 0.36	13.05 $\pm$ 0.33	12.57 $\pm$ 0.72	11.84 $\pm$ 0.59 *
Mean	12.90 $\pm$ 0.19	12.73 $\pm$ 0.11	12.65 $\pm$ 0.32	12.51 $\pm$ 0.13 *	12.39 $\pm$ 0.11 **
DSC (mean $\pm$ std) (%)					
NFL-IPL	92.52 $\pm$ 0.13	92.59 $\pm$ 0.06	92.70 $\pm$ 0.08	92.68 $\pm$ 0.07	92.72 $\pm$ 0.07 *
INL	72.40 $\pm$ 0.53	72.67 $\pm$ 0.29	72.80 $\pm$ 0.68	73.27 $\pm$ 0.41 *	73.24 $\pm$ 0.47 *
OPL-OS	87.50 $\pm$ 0.24	87.51 $\pm$ 0.13	87.54 $\pm$ 0.32	87.59 $\pm$ 0.04	87.66 $\pm$ 0.14
SRF	72.98 $\pm$ 0.55	73.56 $\pm$ 0.56	74.06 $\pm$ 0.83	74.83 $\pm$ 0.60 **	74.82 $\pm$ 0.41 **
RPE	61.98 $\pm$ 0.97	61.85 $\pm$ 0.31	62.42 $\pm$ 0.27	63.22 $\pm$ 0.21	63.45 $\pm$ 0.27 *
PED	68.79 $\pm$ 1.14	68.42 $\pm$ 0.66	69.28 $\pm$ 1.60	69.66 $\pm$ 0.99	70.42 $\pm$ 0.80
IRF	55.91 $\pm$ 3.19	61.99 $\pm$ 2.26 *	63.92 $\pm$ 3.62 *	68.15 $\pm$ 2.18 **	67.98 $\pm$ 2.59 **
Mean	73.15 $\pm$ 0.50	74.08 $\pm$ 0.35 *	74.77 $\pm$ 0.33 **	75.63 $\pm$ 0.39 **	75.76 $\pm$ 0.27 **

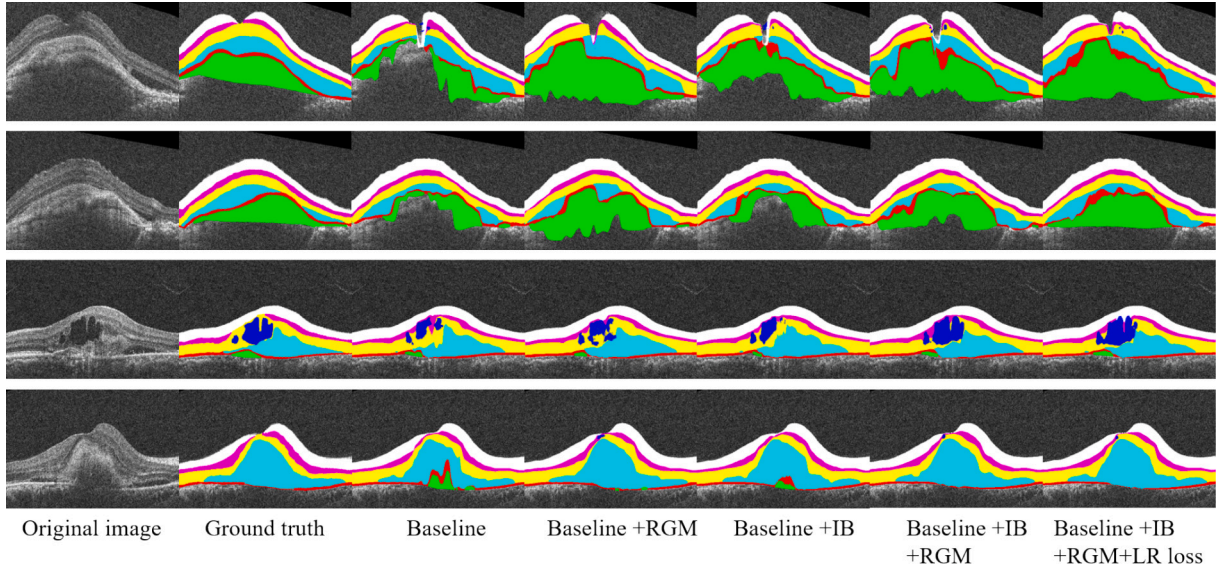


Fig. 6. The visualization results of the ablation studies on the clinical testing set. Color-coded regions represent different retinal layers or fluid lesions.

dataset, including three convolution-based models and three Transformer-based models. The results are shown in Table 3. The convolution-based models showed consistently better performance than Transformer-based models. Specifically, Precat-ResNet50, which is a convolution-based model with residual connections, showed outstanding performance in our experiment. The top three models in terms of MAD were the proposed modified ResNet, Preact-ResNet50, and ConvNext and the top three models in terms of DSC were the proposed modified ResNet, ConvNext, and Preact-ResNet50.

5.4. Comparative study on Duke dataset

We evaluated the proposed framework on the Duke dataset and compared it with the state-of-the-art methods, including Chiu et al. [25], Karri et al. [26], Rathke et al. [27], and He et al. [5].

The results are shown in Table 4. The proposed method outperformed the state-of-the-art methods on most layer-surfaces segmentations. The proposed method showed the best MAD on ILM, NFL-GCL, IPL-INL, ISM-ISE, OS-RPE and BM for patient 1–5, and showed the best

MAD on IPL-INL, INL-OPL, OPL-ONL, ISM-ISE and OS-RPE for patient 6–10.

5.5. Comparative study on RETOUCH dataset

As mentioned above, manual labels were only provided for pathological lesions in this dataset. Thus, the model was first trained on the clinical dataset and cross-dataset learning was applied to generalize the model to the RETOUCH dataset.

First, we compared the performance of the models with and without the proposed cross-dataset learning on the RETOUCH dataset (Fig. 7). It can be seen that directly applying the pre-trained model on the RETOUCH dataset showed degraded performance. There are several reasons for this. First, RETOUCH data are from three vendors with very different image qualities. Second, RETOUCH and our clinical data have different disease distributions; the former contains AMD and RVO patients, while the latter contains only AMD patients. Third, the clinical data consists of typical B-scans with diseases, while RETOUCH consists of 3D volumes with a large proportion of normal B-scans. Thus, the

Table 3
Comparison of different encoder network.

MAD (mean $\pm$ std) (μ m)							
Class	VGG16	Preact-ResNet50	ConvNext	Swin Trans v2	Shuffle Trans	MPViT	Modified ResNet (Proposed)
ILM	5.72 $\pm$ 0.10	5.30 $\pm$ 0.15	5.56 $\pm$ 0.05	5.82 $\pm$ 0.13	5.87 $\pm$ 0.07	5.62 $\pm$ 0.03	5.29 $\pm$ 0.06
IPL-INL	14.95 $\pm$ 0.24	11.01 $\pm$ 0.06	11.40 $\pm$ 0.16	15.05 $\pm$ 0.21	12.16 $\pm$ 0.44	12.10 $\pm$ 0.08	10.36 $\pm$ 0.12
INL-OPL	21.40 $\pm$ 0.16	16.38 $\pm$ 0.32	17.22 $\pm$ 0.19	21.68 $\pm$ 0.35	17.69 $\pm$ 0.64	17.84 $\pm$ 0.38	16.03 $\pm$ 0.38
OuterPRC	16.84 $\pm$ 0.45	12.69 $\pm$ 0.40	13.25 $\pm$ 0.18	15.97 $\pm$ 0.23	13.38 $\pm$ 0.35	13.73 $\pm$ 0.26	11.84 $\pm$ 0.26
IRPE	20.66 $\pm$ 0.52	18.01 $\pm$ 0.70	18.08 $\pm$ 0.82	20.67 $\pm$ 0.64	18.73 $\pm$ 0.24	19.82 $\pm$ 0.60	16.69 $\pm$ 0.14
ORPE	19.51 $\pm$ 0.55	16.28 $\pm$ 0.62	16.47 $\pm$ 0.78	19.36 $\pm$ 0.63	17.36 $\pm$ 0.27	18.08 $\pm$ 0.63	14.67 $\pm$ 0.14
BM	17.16 $\pm$ 0.43	11.46 $\pm$ 0.34	13.16 $\pm$ 0.49	16.93 $\pm$ 0.46	13.57 $\pm$ 0.27	13.72 $\pm$ 0.87	11.84 $\pm$ 0.59
Mean	16.61 $\pm$ 0.21	13.02 $\pm$ 0.21	13.59 $\pm$ 0.27	16.50 $\pm$ 0.25	14.11 $\pm$ 0.22	14.41 $\pm$ 0.33	12.39 $\pm$ 0.11
DSC (mean $\pm$ std) (%)							
NFL-IPL	90.45 $\pm$ 0.09	92.40 $\pm$ 0.04	92.20 $\pm$ 0.09	90.40 $\pm$ 0.08	91.81 $\pm$ 0.21	91.81 $\pm$ 0.05	92.72 $\pm$ 0.07
INL	66.79 $\pm$ 0.45	72.84 $\pm$ 0.27	71.96 $\pm$ 0.33	65.82 $\pm$ 0.28	71.09 $\pm$ 0.90	71.09 $\pm$ 0.41	73.24 $\pm$ 0.47
OPL-OS	83.09 $\pm$ 0.21	87.42 $\pm$ 0.21	86.99 $\pm$ 0.19	83.35 $\pm$ 0.25	86.62 $\pm$ 0.36	86.44 $\pm$ 0.24	87.66 $\pm$ 0.14
SRF	61.38 $\pm$ 1.89	71.83 $\pm$ 0.76	72.78 $\pm$ 0.34	71.40 $\pm$ 0.73	70.44 $\pm$ 0.75	68.01 $\pm$ 1.24	74.82 $\pm$ 0.41
RPE	59.85 $\pm$ 0.66	71.70 $\pm$ 1.40	71.98 $\pm$ 1.08	64.52 $\pm$ 0.87	61.91 $\pm$ 0.40	61.77 $\pm$ 0.34	63.45 $\pm$ 0.27
PED	61.21 $\pm$ 0.65	69.21 $\pm$ 0.85	67.36 $\pm$ 1.44	61.26 $\pm$ 1.10	67.50 $\pm$ 0.55	63.74 $\pm$ 1.73	70.42 $\pm$ 0.80
IRF	28.63 $\pm$ 6.79	54.86 $\pm$ 1.47	58.85 $\pm$ 3.68	36.04 $\pm$ 2.59	59.55 $\pm$ 4.13	55.15 $\pm$ 4.97	67.98 $\pm$ 2.59
Mean	64.49 $\pm$ 1.06	73.02 $\pm$ 0.55	73.17 $\pm$ 0.74	65.83 $\pm$ 0.31	72.70 $\pm$ 0.88	71.15 $\pm$ 0.31	75.76 $\pm$ 0.27

Table 4
Comparison of the proposed method with the state-of-the-art methods on the Duke dataset.

Dataset		MAD (mean $\pm$ std) (μm)							
	Method	ILM	NFL-GCL	IPL-INL	INL-OPL	OPL-ONL	ISM-ISE	OS-RPE	BM
Patient 1–5	Chiu et al. [25]	6.59	8.38	9.04	11.02	11.01	4.84	5.74	5.91
	Karri et al. [26]	4.47	11.77	11.12	17.54	16.74	4.99	5.35	4.30
	Rathke et al. [27]	4.66	6.78	8.87	11.02	13.60	4.61	7.06	5.11
	He et al. [5]	4.51	6.71	8.29	10.71	9.88	4.41	4.52	4.61
	Ours	4.42 $\pm$ 0.12	6.52 $\pm$ 0.25	8.01 $\pm$ 0.13	14.12 $\pm$ 0.23	15.50 $\pm$ 0.35	4.40 $\pm$ 0.18	4.07 $\pm$ 0.08	4.08 $\pm$ 0.04
Patient 6–10	Chiu et al.	5.01	6.37	7.46	7.34	7.74	3.88	4.34	4.32
	Karri et al.	3.64	5.95	6.48	6.64	8.00	3.09	4.12	3.17
	Rathke et al.	3.62	4.87	5.92	7.50	7.69	3.29	4.83	3.16
	He et al.	–	–	–	–	–	–	–	–
	Ours	3.83 $\pm$ 0.14	5.08 $\pm$ 0.12	5.42 $\pm$ 0.14	6.58 $\pm$ 0.16	6.62 $\pm$ 0.09	2.98 $\pm$ 0.11	3.92 $\pm$ 0.11	3.38 $\pm$ 0.08

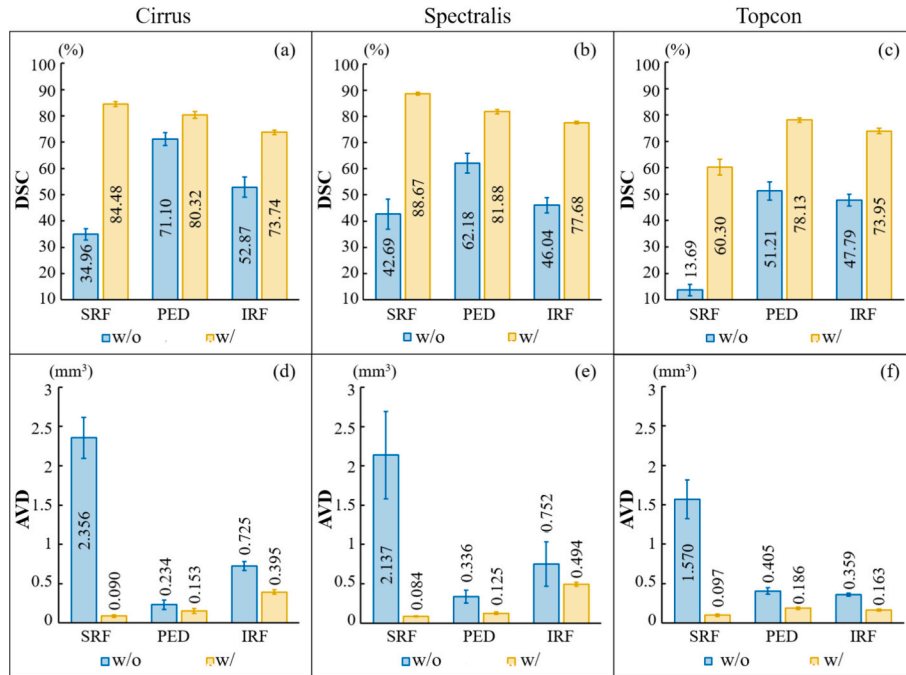


Fig. 7. Comparisons of models with and without cross-dataset learning on the RETOUCH dataset. (a)–(c) and (d)–(f) are the results of DSC and AVD, respectively. The quantitative results of SRF and PED are from the top-level LSR task, and the quantitative results of IRF are from the independent IRF branch.

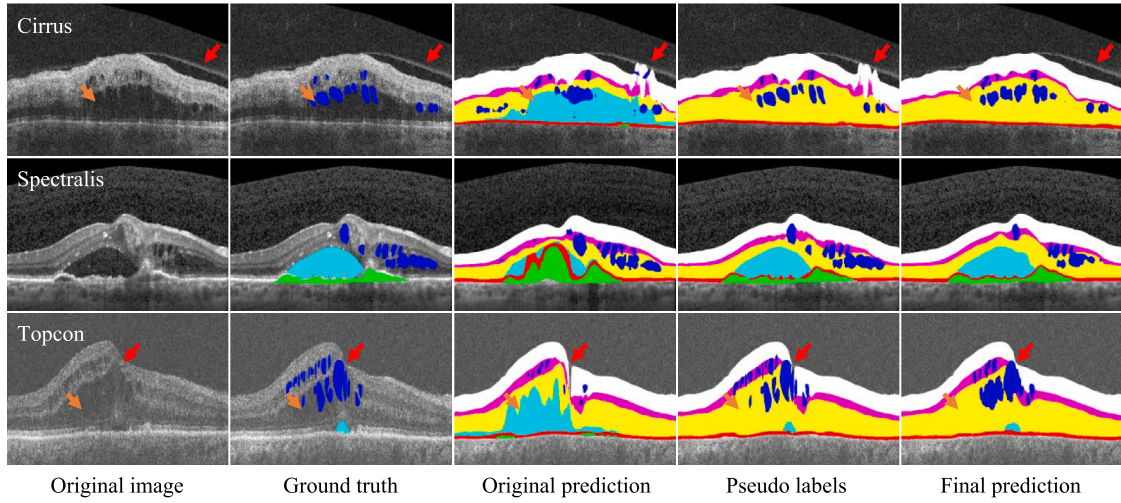


Fig. 8. Typical results on the RETOUCH training set. Different colored areas represent different retinal layers or fluids. The results of retinal layers, SRF and PED are from the top-level LSR task, and the results of IRF are from the independent IRF branch.

original prediction without cross-dataset learning contains lots of false positives, which is reflected by the large AVD values and erroneous detected regions.

The model with cross-dataset learning showed significant improvements, with steadily increased DSC and decreased AVD for all types of fluid lesions on all vendors. For example, DSC of Cirrus SRF increased from 34.96 % to 84.48 %, and AVD decreased from 2.356 mm³ to 0.090 mm³.

The visualized original prediction, corrected pseudo labels, and final prediction are shown in Fig. 8. The initial prediction contained erroneous segmentations for both retinal layers and fluid lesions. After correcting the pseudo labels, fluid lesion segmentation showed obvious improvements. It is noteworthy that the original prediction also contained some layer regression errors (red arrow), which could also be corrected in the final prediction after correcting the pseudo labels.

At last, we compared the pathological lesion segmentation with the literatures (Table 5). The proposed method showed comparable DSC with the literature methods on all vendors. Meanwhile, the proposed method showed significantly better AVD on all vendors and fluid types except the PED on Spectralis. The proposed method showed comparable DSC but significantly better AVD, indicating our method gave a better performance on large lesions over small lesions.

6. Discussion and conclusion

The complex interaction between retinal layers and fluid lesions has drawn the attention of researchers and a number of studies have proposed various types of constraints to improve the performance. Topological constraints have been applied between layers, from layer to fluid,

and between layers and fluid lesions. Some studies applied a sequential segmentation strategy to enforce the constraints. Constraints can also be implemented through post-processing. For example, Montuoro used an auto-context loop method to iteratively include spatial context extracted from previous classifications to refine the prediction result in the next iteration [16]. These methods implement one-way constraints and a succeeding task cannot have an influence on a preceding task. In this study, we propose an end-to-end hierarchical architecture with asymmetric information flow to achieve topological constraints between layers and between layers and fluid lesions. Specifically, LCM provides hard constraint between layers. RGM is an attention gate consisting of both hard spatial attention and channel attention.

The model complexity is evaluated by analyzing the number of parameters (Params). The Params for one-, two-, and three-branch architectures (i.e., decoder 1, decoder 1–2, decoder 1–3) are 28.0 M, 31.0 M, and 35.8 M, respectively. As the number of branches increases, the model's ability to capture complex relationships between subtasks improves, but this also leads to higher computational demands. Specifically, compared to the single-branch model, the three-branch model has a 27.8 % increase in model complexity. Nevertheless, considering the lightweight model backbone, the overall model complexity remains manageable.

There are several limitations of the proposed study. In this study, we focus on the joint segmentation of retinal layers and fluid lesions. Hyper-reflective lesions (e.g., subretinal hyperreflective material), which are also important biomarkers for the diagnosis and assessment of AMD, are not considered. Second, balancing different tasks in the loss function is challenging. Currently, we empirically adjust the weights of different tasks to balance their contribution to the final loss. Several approaches

Table 5

Comparison of the proposed method with other methods on the RETOUCH dataset.

Method	DSC (mean $\pm$ std) (%)								
	Cirrus			Spectralis			Topcon		
	SRF	PED	IRF	SRF	PED	IRF	SRF	PED	IRF
Lu et al. [28]	75.0	72.9	75.3	87.3	73.7	74.0	64.3	69.7	66.5
Xing et al. [12]	82.8 $\pm$ 3.1	81.5 $\pm$ 3.5	70.7 $\pm$ 0.7	90.7 $\pm$ 0.9	77.5 $\pm$ 0.3	74.9 $\pm$ 0.5	68.5 $\pm$ 2.2	79.5 $\pm$ 2.8	72.9 $\pm$ 0.3
Ours	84.5 $\pm$ 0.9	80.3 $\pm$ 1.3	73.7 $\pm$ 0.8	88.7 $\pm$ 0.4	81.9 $\pm$ 0.9	77.7 $\pm$ 0.4	60.3 $\pm$ 3.0	78.1 $\pm$ 0.8	74.0 $\pm$ 1.0
Method	AVD (mean $\pm$ std) (mm ³)								
	Cirrus			Spectralis			Topcon		
	SRF	PED	IRF	SRF	PED	IRF	SRF	PED	IRF
Lu et al.	—	—	—	—	—	—	—	—	—
Xing et al.	0.50 $\pm$ 1.70	0.42 $\pm$ 0.46	0.99 $\pm$ 0.47	0.32 $\pm$ 0.13	0.01 $\pm$ 0.01	0.95 $\pm$ 0.19	0.19 $\pm$ 0.19	0.41 $\pm$ 0.19	1.27 $\pm$ 0.37
Ours	0.09 $\pm$ 0.02	0.15 $\pm$ 0.03	0.40 $\pm$ 0.03	0.08 $\pm$ 0.01	0.13 $\pm$ 0.02	0.49 $\pm$ 0.03	0.10 $\pm$ 0.02	0.19 $\pm$ 0.02	0.16 $\pm$ 0.02

have been explored to address the multitasking optimization problem, including methods based on dynamically adjusting the loss weight and methods based on adjusting the gradient value and direction between different tasks. The future work will focus on automatically optimizing the loss function design of the multi-task architecture.

In this study, we propose a three-branch, hierarchical multi-task framework that enables a joint regression of seven retinal layers and segmentation of three types of pathological lesions. We also propose a cross-dataset learning strategy to leverage the public datasets with partial labels. The proposed framework was comprehensively evaluated on a private clinical dataset and two widely used public datasets. Ablation and comparative studies showed that the hierarchical multi-task architecture significantly improved the segmentation accuracy and outperform the state-of-the-art methods. The proposed framework has the potential to be generalized to other image segmentation tasks that have interdependent relationships.

CRedit authorship contribution statement

Xiayu Xu: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Hualin Wang:** Data curation. **Yulei Lu:** Data curation. **Hanze Zhang:** Data curation. **Tao Tan:** Writing – original draft, Methodology. **Feng Xu:** Project administration. **Jianqin Lei:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The clinical dataset and manual labels are publicly available on GitHub (<https://github.com/xjtu-mia/OCT-Multitask>).

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